Preprint

SUPPLEMENTARY MATERIAL The Locked Bimetric Potential and the Constitutive Law

Stephen Nelson Field Dynamics Research

stephen@fielddynamics.org

Abstract

Papers VIII, IX and XII fix the Hassan–Rosen coefficients to $\beta = (1, 0, 16, 0, 1)$ and state that the locked bimetric theory and the SST are two projections of one geometry, so that the bimetric theory's quasi-static limit should give $\mu(x) = x/(x+1)$. We settle what can and cannot be shown. *Negative:* in the standard reduction (decoupling limit, static spherical weak field) the response depends on the enclosed mass only through M/r^3, for every β; no function of M/r^3 equals a function of $x \propto M/r^2$ for all masses and radii; the enhancement is bounded, $g_{\rm tot}/g_N < 1 + 1/\sqrt{2} < 1.71$; and the potential vanishes at GFD's parity configuration. So the standard reduction cannot give μ, and the obstruction is structural, not specific to the lock. *Positive:* on the frame deformed along a single leg by a ratio w, every potential is $V = A + Bw$, and the deformed leg's share $Bw/(A+Bw)$ obeys exchange closure $s(1/w) = 1 - s(w)$ if and only if $\beta_0 + 2\beta_1 = \beta_4 + 2\beta_3$. It is then exactly $w/(1+w)$. Palindromic coefficients (the $g \leftrightarrow f$ symmetry) satisfy the condition, the locked set gives $V = 49(1+w)$, and $\beta_2 = 16$ cancels. With $w = |a|/a_0$, a the acceleration of the tetrad's time leg, this is $\mu(x)$. The deformation is not a Hassan–Rosen reference metric, and the energy F is not supplied by the potential. The correct statement is therefore: *an exchange-symmetric bimetric potential, evaluated on the frame deformed along the time leg's acceleration, carries the face share $x/(x+1)$; the standard Hassan–Rosen reduction does not.*

Keywords: Hassan–Rosen, bimetric gravity, Vainshtein, Galileon, exchange symmetry, constitutive law, MOND, proof

Setting

The Hassan–Rosen (HR) interaction is $m^2\sqrt{-g}\,V(S)$, $V(S) = \sum_{n=0}^{4}\beta_n e_n(S)$, with $S = \sqrt{g^{-1}f}$ and e_n the elementary symmetric polynomials of its eigenvalues [1]. The lock is $\beta = (1, 0, 16, 0, 1)$ [2, 3]. The SST's scalar sector is $-(a_0^2/8\pi G)F(y)$, $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$, $\mu(x) = F'(x^2) = x/(x+1)$ [4, 5]. The question, listed as open in Papers VIII and IX, is whether the first reduces to the second. Each step is checked by the accompanying script, 18 checks, exact algebra [9]; the decoupling-limit derivation is in [8] (23 checks).

Part A: the standard reduction

Theorem 1 (the response depends on density). *In the decoupling limit, with Stückelberg field π and $\Lambda_3^3 = m^2 M_{\rm Pl}$, the h–π mixing has coefficients $d_0 = \beta_0 + 3\beta_1 + 3\beta_2 + \beta_3$, $d_1 = \beta_1 + 2\beta_2 + \beta_3$, $d_2 = \beta_2 + \beta_3$, $d_3 = \beta_3$; for the lock, $(49, 32, 16, 0)$. Around a static mass, with $d_3 = 0$, $\lambda = \pi'/r = -sq/(2Ma_1)$ and $Q = a_2 q/(Ma_1^2)$, the π-equation is*

$$Q^2 s^3 + 6Qs^2 + 6s - 2 = 0, \qquad g_{\rm tot}/g_N = 1 + s, \tag{1}$$

with $q = M_{\rm enc}/4\pi r^3$. The radius enters only through q.

Proof. The coefficients follow from expanding $\sum(-1)^n\beta_n Y_n(1-\Pi)$ in the identically conserved basis; the reduced first integrals are in [8]. The substitution is checked exactly. □

$d_1 = 32$ is the Fierz–Pauli combination and $d_3 = 0$ removes the quintic mixing; $d_2 = 16$ keeps the cubic Galileon term (Paper XII states that both vanish).

Theorem 2 (bounded enhancement). *As $Q \to 0$, $s = 1/3$: $g_{\rm tot}/g_N = 4/3$ (vDVZ). As $Q \to +\infty$, $s \to 0$: screened to general relativity. On the $Q < 0$ branch the turning point is $Q_* = 2 - 2\sqrt{2}$, where $s = 1/\sqrt{2}$. So $g_{\rm tot}/g_N < 1.71$ on every branch.*

Proof. With $v = Qs$, (1) is $v(v^2 + 6v + 6) = 2Q$, whose turning points are $v = -2 \pm \sqrt{2}$; all values are checked exactly. □

Theorem 3 (no change of variable). *No function of M/r^3 equals a function of $x = GM/(a_0 r^2)$ for all M and r, unless both are constant. Hence the standard response cannot be $1/\mu(x)$, for any β.*

Proof. Along $M = Lr^3$ the density is fixed while $x = GLr/a_0$ varies, and $\mathrm{d}[1/\mu(x)]/\mathrm{d}r \neq 0$ (checked). Equivalently, the Vainshtein radius scales as $M^{1/3}$ and the MOND radius $\sqrt{GM/a_0}$ as $M^{1/2}$. □

Flat rotation curves need $g_{\rm tot}/g_N \to \infty$ as $g_N \to 0$, which Theorem 2 excludes. With $m \sim H_0$, galaxies also lie well inside the Vainshtein radius (∼260 kpc at $10^{10} M_\odot$), where

$\mu_{\rm eff} \to 1$ [8]. Dimensionally, the Galileon variable is $\Phi/(mr)^2$ and the MOND variable $\Phi/(mr)$: both small quantities Φ/c^2 and rH_0/c are $\sim 10^{-6}$ at galactic radii, which is why $a_0 \sim cH_0$ is natural, but the potential's strongest interactions select the wrong power.

Theorem 4 (the parity configuration and the polynomial). *At* $S = P = \mathrm{diag}(1,-1,-1,-1)$, $e_n(P) = (1,-2,0,2,-1)$ *and* $V(P) = \beta_0 - \beta_4 = 0$. *On* $S = \mathrm{diag}(1,s,s,s)$ *the locked potential is* $1 + 48s + 48s^2 + s^3$ *(98, 1, 0 at* $s = +1, 0, -1$*), not the coupling polynomial* $1 + 4s + 16s^2$ *(21, 17, 13).*

Proof. Direct evaluation (checked). □

So GFD's two roots (one metric, $S = P$) and HR's two metrics are different objects, and the potential's values are not the polynomial's.

Part B: the acceleration-deformed frame

Theorem 5 (exchange closure fixes the share). *On* $S = \mathrm{diag}(1,1,1,w)$*, for every* β*,*

$$V = A + Bw,$$
$$A = \beta_0 + 3\beta_1 + 3\beta_2 + \beta_3,$$
$$B = \beta_1 + 3\beta_2 + 3\beta_3 + \beta_4.$$

The deformed leg's share $s(w) = Bw/(A + Bw)$ *satisfies* $s(1/w) = 1 - s(w)$ *if and only if* $A = B$*, that is*

$$\beta_0 + 2\beta_1 = \beta_4 + 2\beta_3, \tag{2}$$

and then $s(w) = w/(1 + w)$ *exactly.*

Proof. $s(1/w) - [1 - s(w)] = -w(A - B)(A + B)/[(A + Bw)(Aw + B)]$ (checked), which vanishes iff $A = B$ since $A + B > 0$. □

Corollary 1 (the lock and β_2). *Palindromic* β*, equivalent to HR's* $g \leftrightarrow f$ *symmetry* $\det S\, V(S^{-1}) = V(S)$*, satisfies (2), and then* $V = (\beta_0 + 4\beta_1 + 3\beta_2)(1 + w)$*. The locked set gives* $V = 49(1 + w)$*, and* β_2 *cancels from* s*. On this family* $V/V(I) = \det[(I + S)/2]$*, the volume of the midpoint frame.*

Proof. All identities checked exactly; $(1, 0, 16, 0, 2)$ shows that (2) is a real condition. □

This is the exchange-closure theorem of the constitutive law [7] at the level of the bimetric potential: the share's form follows from the $g \leftrightarrow f$ symmetry, not from the value 16.

Theorem 6 (the time leg, and what the potential does not supply). *(i) With* $w = |a|/a_0$*, a the acceleration of the tetrad's time leg, a static field has* $c^2|a| = |\nabla\Phi|$ *exactly [6], so* $w = x$ *and* $s = \mu(x)$*. (ii) The auxiliary energy*

$$E(u; y) = (1 - e^{-u})\, y - 2(\sinh u - u),$$

with $u = \ln(V/V_0)$*, is stationary at* $V/V_0 = 1 + \sqrt{y}$*, where its value is* $F(y)$ *and its slope* $\partial E/\partial y$ *is* μ*. (iii) Its* u*-dependence contains no* β*.*

Proof. All checked exactly. □

So the potential supplies the variable $V/V_0 = 1 + x$; the energy F, and with it the dynamics, comes from the SST's own scalar term.

Conclusion and scope

- The locked HR theory does not reduce to $\mu(x)$ by the standard reduction (Theorems 1–3), and no choice of β would change that. Papers VIII, IX and XII overstate the equivalence.
- What the lock does fix is bimetric physics: the de Sitter background $\Lambda = m^2(1 + 48c^2)$, the Fierz–Pauli mass, the Higuchi bound and the screening cubic (1) [8].
- The constitutive law follows from the potential's exchange symmetry on the frame deformed along the time leg's acceleration (Theorem 5, Corollary 1, Theorem 6). That frame is not an HR reference metric; it is the single-metric time-leg structure of [6].
- Not done here: the full bimetric (dynamical f) decoupling limit, which keeps the linear enhancement below 4/3; and a derivation of the deformation $w = |a|/a_0$ from an action.

References

[1] S. F. Hassan and R. A. Rosen, Bimetric gravity from ghost-free massive gravity, *JHEP* **02** (2012) 126.

[2] S. Nelson, *GFD and Bimetric Gravity*, Paper VIII (2026).

[3] S. Nelson, *Dual-tetrad covariance and the Hassan–Rosen lock*, Paper XII (2026).

[4] S. Nelson, *Field Closure of the Tetrad Frame*, Letter 3 (2026).

[5] S. Nelson, *The SST Covariant Completion in Its Quasi-Static Limit* (2026), `proofs/sst-quasi-static-limit/`.

[6] S. Nelson, *The SST Covariant Completion: y From the Acceleration of the Time Leg* (2026), `proofs/sst-covariant-time-leg/`.

[7] S. Nelson, *The constitutive law from exchange symmetry* (2026), `proofs/constitutive-law-exchange-symmetry/`.

[8] S. Nelson, *Does the locked bimetric potential project onto $\mu(x)$?* (2026), `research/bimetric-projection/`.

[9] S. Nelson, `verify_bimetric_lock_constitutive_law.py` (2026), `proofs/bimetric-lock-constitutive-law/`.